

Landing Zone

- **Purpose:** A Landing Zone is a **ready-made and secure AWS environment** that provides a strong foundation for an organization to start using AWS.
 - **Key Points:**
 - o Creates a proper **AWS account structure** for different teams, projects, and environments.
 - o Keeps important resources like **security, networking, and logging** organized.
 - o Provides a **secure and controlled environment** for running applications.
 - o Helps maintain the **same security and configuration standards** across AWS accounts.
 - o Uses AWS Organizations to **manage multiple AWS accounts centrally**.
 - o Helps control who can **access and use AWS resources**.
 - o Makes it easier to **add new AWS accounts** as the organization grows.
 - o Helps organizations follow **security, compliance, and governance rules**.
 - o Acts as the **foundation on which AWS applications and services are built**.

AWS Control Tower

- **Purpose:** AWS Control Tower is a service that **automatically sets up and manages a secure AWS environment with multiple accounts**.
 - **Key Points:**
 - o Helps create a **well-organized AWS environment** quickly.
 - o Automatically sets up and manages the **Landing Zone**.
 - o Uses **guardrails** to make sure AWS accounts follow company security and compliance rules.
 - o Makes it easier to **create and manage new AWS accounts**.
 - o Provides a **central dashboard** to monitor accounts and their compliance status.
 - o Helps detect when an account **does not follow the required rules**.
 - o Provides centralized management of **users and access**.
 - o Reduces the amount of **manual work** required to manage multiple AWS accounts.
 - o Helps organizations maintain **security, consistency, and governance** as their AWS environment grows.

Easy way to remember

Landing Zone = The Foundation 🏗️

It creates the **proper structure and secure environment** for your AWS organization.

Control Tower = The Manager 👮

It **sets up, monitors, and manages** that environment and makes sure everyone follows the rules.

Simple example:

Imagine a company is building a large office campus:

- **Landing Zone** → Designs the campus with separate buildings for HR, Finance, IT, Security, etc.

- **Control Tower** → Manages the campus, controls access, checks the rules, and makes sure every building follows company policies.

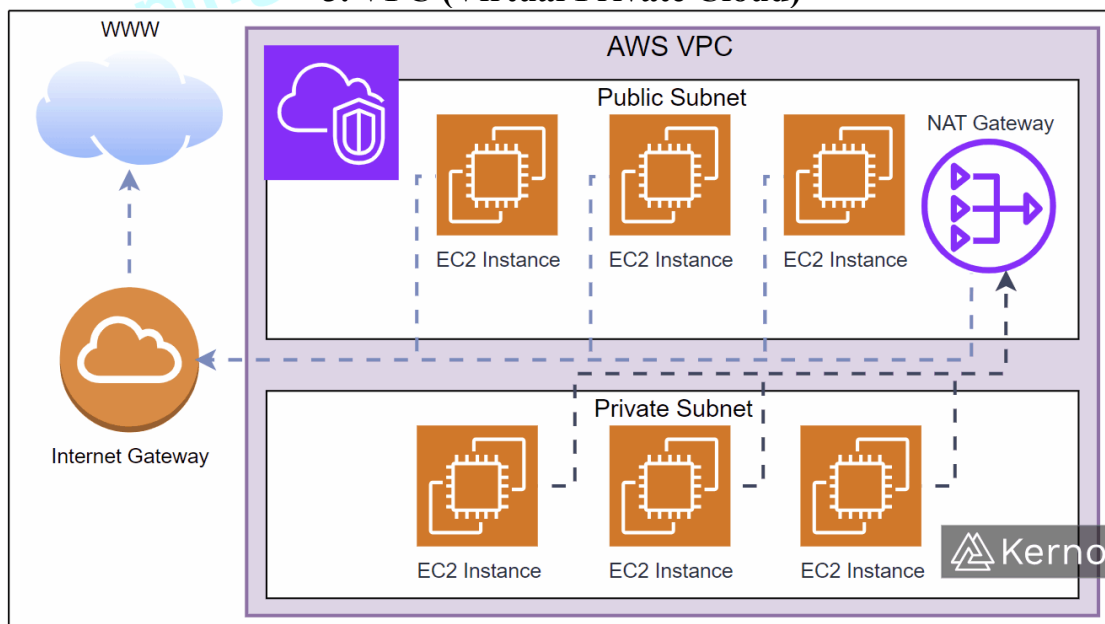
1. EC2 (Elastic Compute Cloud)

- **Purpose:** Scalable virtual servers in the cloud for running applications.
- **Key Points:**
 - Customizable configurations (CPU, memory, storage, networking).
 - Flexible pricing models: On-demand, Reserved, and Spot Instances.
 - Auto Scaling for dynamic resource adjustment based on traffic.
 - Supports Linux, Windows, and custom AMIs.
 - Integrates with VPC, S3, and IAM for better security and networking.

2. IAM (Identity and Access Management)

- **Purpose:** Secure management of access to AWS resources.
- **Key Points:**
 - Creates users, groups, roles, and policies to control permissions.
 - Supports MFA (Multi-Factor Authentication) for enhanced security.
 - Fine-grained access control for different AWS resources.
 - Centrally manages and audits user activities.
 - Works with AWS services to manage permissions across all resources.

3. VPC (Virtual Private Cloud)



- **Purpose:** Creates isolated networks in AWS for resources.

- **Key Points:**
 - Customizable IP ranges, subnets, and route tables.
 - Provides secure resource communication with security groups and NACLs.
 - Supports VPN and Direct Connect for hybrid cloud setups.
 - Multi-AZ deployment for high availability.
 - Ensures fault tolerance and security for resources.
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4. S3 (Simple Storage Service)

- **Purpose:** Scalable object storage for storing and retrieving data.
 - **Key Points:**
 - Virtually unlimited, durable, and cost-effective storage.
 - Multiple storage classes for varying access needs.
 - Built-in encryption for data at rest and in transit.
 - Versioning for object management and restoration.
 - Easily integrates with Lambda, CloudFront, and other AWS services.
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5. Session Manager

- **Purpose:** Secure shell access to EC2 instances without needing SSH/RDP.
 - **Key Points:**
 - Remote management without the need for opening inbound ports.
 - Logging and auditing for session activities.
 - Allows access to EC2 instances in private subnets.
 - Integrates with AWS Systems Manager for automation.
 - No need for public IP addresses on instances.
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6. Route 53

- **Purpose:** DNS service for routing traffic to resources.
 - **Key Points:**
 - Domain registration and DNS routing management.
 - Configures routing policies like weighted, latency-based, and geolocation.
 - Health checks and DNS failover to improve availability.
 - Integrates with CloudFront and ELB for dynamic routing.
 - Offers fast DNS resolution for applications.
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7. CloudTrail

- **Purpose:** Monitors and records AWS API calls for auditing.

- **Key Points:**
 - Tracks changes made to AWS resources and user activities.
 - Logs are stored in S3 for security and analysis.
 - Helps with compliance and auditing for regulatory requirements.
 - Integrates with CloudWatch for real-time monitoring.
 - Provides detailed logs for troubleshooting security incidents.
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8. CloudWatch

- **Purpose:** Monitoring and observability service for AWS resources.
 - **Key Points:**
 - Collects logs, metrics, and events from AWS resources.
 - Set up alarms to trigger automated actions or notifications.
 - Helps visualize resource performance and system health.
 - Allows for centralized log storage and analysis.
 - Integrates with Auto Scaling and Lambda for automated management.
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9. EKS (Elastic Kubernetes Service)

- **Purpose:** Managed Kubernetes service for containerized applications.
 - **Key Points:**
 - Automates the deployment and scaling of containerized applications.
 - Provides integration with AWS services like IAM, VPC, and CloudWatch.
 - Supports multi-cluster and multi-region architectures.
 - Built-in security features like IAM roles for service accounts.
 - Integrates easily with DevOps workflows and CI/CD pipelines.
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10. Auto Scaling

- **Purpose:** Automatically adjusts resources to meet demand.
 - **Key Points:**
 - Scales EC2 instances, and other resources based on real-time needs.
 - Ensures cost-efficiency and high availability of applications.
 - Works with Elastic Load Balancer (ELB) to balance traffic.
 - Supports scaling across multiple availability zones.
 - Provides automated scaling for ECS and Lambda functions.
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11. Lambda

- **Purpose:** Serverless compute service that runs code in response to events.
- **Key Points:**

- No need to provision or manage servers.
 - Supports multiple languages like Node.js, Python, Java, and Go.
 - Pay only for the compute time your code uses.
 - Can be triggered by events from S3, DynamoDB, SNS, API Gateway, and more.
 - Ideal for event-driven architectures and microservices.
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12. AWS Shield

- **Purpose:** DDoS protection for AWS services.
 - **Key Points:**
 - Shield Standard automatically protects AWS customers.
 - Shield Advanced offers enhanced protection, including real-time attack visibility.
 - Protects resources like CloudFront, Route 53, and ELB.
 - Provides 24/7 DDoS response team (DRT) support.
 - Helps mitigate large-scale attacks without affecting performance.
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13. AWS WAF (Web Application Firewall)

- **Purpose:** Protects web applications from common security threats.
 - **Key Points:**
 - Customizable rules to filter malicious web traffic.
 - Protects against SQL injection, cross-site scripting, and bots.
 - Integrates with CloudFront, ALB, and API Gateway.
 - Provides real-time logging of web traffic patterns.
 - Automates security responses using AWS Lambda.
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14. Amazon GuardDuty

- **Purpose:** Threat detection service for AWS environments.
 - **Key Points:**
 - Continuously monitors AWS accounts for suspicious activity.
 - Uses machine learning and threat intelligence to detect anomalies.
 - Provides actionable findings to address security risks.
 - Integrates with CloudWatch for automatic response actions.
 - Helps identify unauthorized access, reconnaissance, and data exfiltration.
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15. Amazon Inspector

- **Purpose:** Automated vulnerability management service for AWS workloads.

- **Key Points:**

- Continuously scans AWS workloads (EC2 instances, container images, and Lambda functions) for security vulnerabilities.
- Automatically discovers resources and assesses them against known CVEs and security best practices.
- Provides detailed findings with severity ratings and remediation guidance.
- Integrates with Amazon EventBridge and CloudWatch for automated alerts and response workflows.
- Helps identify software vulnerabilities, unintended network exposure, and deviations from security best practices.

16. AWS Key Management Service (KMS)

- **Purpose:** Manages encryption keys for securing data within AWS.

- **Key Points:**

- Supports symmetric and asymmetric encryption keys.
- Easily integrates with AWS services like S3, RDS, and EBS.
- Automates key rotation to simplify key management.
- Provides fine-grained access control for encryption keys.
- Audits key usage via CloudTrail for compliance.

17. AWS Systems Manager (SSM)

- **Purpose:** Automates operational tasks and manages AWS infrastructure.

- **Key Points:**

- Manages patching, configurations, and resource compliance.
- Supports remote management of EC2 instances and on-premises resources.
- Automates routine administrative tasks like software deployments.
- Provides centralized management with CloudWatch and CloudTrail.
- Helps with security and compliance auditing.

18. Amazon RDS (Relational Database Service)

- **Purpose:** Managed relational database service for multiple database engines.

- **Key Points:**

- Supports MySQL, PostgreSQL, Oracle, SQL Server, and MariaDB.
- Automates backups, patch management, and failover.
- Provides scalability through read replicas and multi-AZ deployments.
- Offers encryption at rest and in transit for secure data management.
- Integrates with CloudWatch for monitoring and performance metrics.

19. Amazon Aurora

- **Purpose:** High-performance relational database compatible with MySQL and PostgreSQL.
- **Key Points:**
 - Provides up to 5x the performance of MySQL and 2x that of PostgreSQL.
 - Self-healing storage and automated backups.
 - Aurora Serverless scales automatically based on demand.
 - Multi-AZ deployments for high availability.
 - Fully managed with automatic patching and scaling.

20. Amazon DynamoDB

- **Purpose:** Managed NoSQL database service for high-performance applications.
- **Key Points:**
 - Provides fast, low-latency read and write operations.
 - Automatically scales throughput and storage.
 - Supports both key-value and document data models.
 - Built-in encryption at rest for security.
 - Integrates with Lambda, CloudWatch, and other AWS services.

21. Amazon Redshift

- **Purpose:** Managed data warehouse service for running complex queries.
- **Key Points:**
 - Provides fast SQL-based querying for structured data.
 - Columnar storage and parallel query processing.
 - Scales with Redshift Spectrum for querying data in S3.
 - Integrates with AWS analytics services like Glue and EMR.
 - Offers encryption for data security.

22. AWS DMS (Database Migration Service)

- **Purpose:** Helps migrate databases to AWS with minimal downtime.
- **Key Points:**
 - Supports migrations between homogeneous and heterogeneous databases.
 - Continuous data replication to ensure data integrity.
 - Monitors migration progress and provides detailed reports.
 - Enables migration of on-premises databases to AWS.

- Supports many database engines, including Oracle, MySQL, and SQL Server.
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23. AWS Batch

- **Purpose:** Efficiently runs batch processing workloads at scale.
 - **Key Points:**
 - Automatically provisions the compute resources for batch jobs.
 - Handles thousands of batch computing jobs at any scale.
 - Supports custom compute environments and job scheduling.
 - Integrates with S3 for data storage and processing.
 - Supports EC2 Spot Instances for cost optimization.
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24. AWS Glue

- **Purpose:** Managed ETL service for data preparation and loading.
 - **Key Points:**
 - Automatically crawls and discovers datasets.
 - Transforms, cleans, and prepares data for analytics.
 - Integrates with Amazon Redshift, S3, and other data sources.
 - Supports both Python and Scala for scripting.
 - Serverless architecture removes the need for infrastructure management.
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25. Amazon Elastic File System (EFS)

- **Purpose:** Scalable file storage service for EC2 instances and on-premises resources.
 - **Key Points:**
 - Provides scalable, shared file storage accessible by EC2 instances.
 - Fully managed with automatic scaling.
 - Supports NFS protocols for integration with on-premises environments.
 - Provides encryption at rest and in transit.
 - Ideal for use cases like content management and big data analytics.
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26. AWS Elastic Beanstalk

- **Purpose:** Platform-as-a-Service (PaaS) for deploying and managing applications.
- **Key Points:**
 - Supports multiple programming languages, including Java, Python, Node.js, and more.
 - Automatically handles deployment, scaling, and monitoring.
 - Simplifies application management with an easy-to-use console.

- Integrates with other AWS services like RDS, S3, and SNS.
 - Allows easy customization with configuration options.
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27. Amazon CloudFront

- **Purpose:** Content delivery network (CDN) for faster content delivery.
 - **Key Points:**
 - Distributes content to users globally with low latency.
 - Supports dynamic and static content delivery.
 - Can serve from edge locations close to users for faster speeds.
 - Integrates with S3, EC2, and Lambda for dynamic content delivery.
 - Provides DDoS protection and encryption for security.
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28. AWS Snowball

- **Purpose:** Data transport solution for securely transferring large amounts of data to AWS.
 - **Key Points:**
 - Provides a physical appliance to transport data to AWS.
 - Useful for large-scale data migrations when internet transfer is impractical.
 - Integrates with S3 and other storage services for data import.
 - Supports encryption for secure data handling.
 - Offers Snowball Edge for local compute and storage capabilities.
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29. AWS CodePipeline

- **Purpose:** Continuous integration and delivery service for automated software workflows.
 - **Key Points:**
 - Automates the build, test, and deployment process of applications.
 - Integrates with GitHub, CodeCommit, and other source control repositories.
 - Supports multiple stages for flexible pipeline configurations.
 - Can integrate with Lambda for serverless application deployments.
 - Provides full visibility into your CI/CD process.
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30. AWS CodeBuild

- **Purpose:** Managed build service to compile source code and run tests.
- **Key Points:**
 - Compiles code from source control like GitHub and CodeCommit.
 - Runs tests on your code and produces build artifacts.

- Scales automatically based on build volume.
 - Integrates with AWS CodePipeline and other CI/CD tools.
 - Supports multiple build environments like Java, Node.js, Python, and Go.
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31. AWS CodeDeploy

- **Purpose:** Automates code deployment to EC2 instances and on-premises servers.
 - **Key Points:**
 - Supports rolling, blue/green, and canary deployments.
 - Integrates with AWS services like EC2, Lambda, and ECS.
 - Provides detailed deployment logs for visibility.
 - Enables zero-downtime application deployments.
 - Supports integration with third-party CI/CD tools.
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32. AWS X-Ray

- **Purpose:** Tracing service for debugging and analyzing microservices applications.
- **Key Points:**
 - Provides insights into the performance of distributed applications.
 - Helps identify performance bottlenecks and troubleshoot errors.
 - Supports integration with Lambda, EC2, and other services.
 - Visualizes application architecture and request paths.
 - Helps with security and compliance monitoring.

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